

GRAND STRAND, SC, USA

WMO: 747915

Lat: 33.816N

Lon: 78.720W

Elev: 10

StdP: 101.21

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93718

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.1	-13.8	1.1	0.0	-11.1	1.4	2.1	9.9	15.5	8.7	13.9	2.0	20	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	32.5	25.6	31.4	25.5	30.6	25.3	27.4	30.4	26.8	29.9	26.3	29.4	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.2	25.9	21.3	29.0	25.2	20.4	28.6	86.2	30.5	83.8	29.9	81.5	29.3	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.2	7.4	-7.2	35.2	1.8	2.2	-8.5	36.8	-9.5	38.1	-10.5	39.3	-11.8	40.9
			-8.5	28.1	1.6	0.5	-9.7	28.5	-10.6	28.8	-11.6	29.1	-12.7	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.7	9.2	12.6	17.0	21.5	25.5	26.8	26.5	24.2	18.9	13.3	10.0	(6)
(7)		DBStd	7.79	5.39	4.46	4.51	3.73	3.14	2.23	1.79	1.79	2.51	4.08	4.53	5.01	(7)
(8)		HDD10.0	287	110	63	27	2	0	0	0	0	1	18	65		(8)
(9)		HDD18.3	1306	328	255	183	66	10	0	0	1	43	158	261		(9)
(10)		CDD10.0	3148	40	42	107	213	358	466	520	513	427	278	119	65	(10)
(11)		CDD18.3	1124	0	1	4	27	109	216	262	254	178	62	8	2	(11)
(12)		CDH23.3	9916	2	1	13	78	616	2023	2759	2601	1436	367	18	1	(12)
(13)		CDH26.7	2843	1	0	2	13	93	539	929	867	359	41	1	0	(13)
(14)	Wind	WSAvg	3.3	3.0	3.3	3.6	3.9	3.8	3.7	3.6	3.2	3.3	2.9	2.8	2.9	(14)
(15)	Precipitation	PrecAvg	1295	90	89	96	70	93	121	147	170	159	103	72	84	(15)
(16)		PrecMax	1780	223	304	206	166	182	322	264	360	544	343	151	257	(16)
(17)		PrecMin	870	21	24	29	3	32	10	77	85	40	3	4	25	(17)
(18)		PrecStd	211	50	54	45	44	38	66	49	64	112	83	38	51	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	22.3	24.9	28.4	31.7	33.9	34.4	33.7	31.7	29.4	25.8	22.6	(19)
(20)			MCWB	16.7	16.2	16.8	18.8	23.5	25.1	26.0	26.7	24.2	23.4	21.1	19.1	(20)
(21)		2%	DB	19.0	19.6	22.5	25.1	28.8	31.8	32.5	32.0	30.3	27.7	23.4	21.0	(21)
(22)			MCWB	16.0	14.7	16.9	19.5	22.5	25.0	26.1	26.2	24.2	22.6	19.1	18.2	(22)
(23)		5%	DB	17.6	17.9	21.0	23.8	27.4	30.4	31.3	31.1	29.5	26.7	22.3	19.5	(23)
(24)			MCWB	15.2	14.6	16.5	19.0	22.4	24.8	26.0	25.9	24.1	21.9	18.6	17.3	(24)
(25)		10%	DB	16.2	16.4	19.2	22.7	26.3	29.2	30.5	30.2	28.7	25.6	21.0	17.9	(25)
(26)			MCWB	14.0	14.0	15.7	18.5	21.9	24.4	25.6	25.4	23.8	21.4	17.9	16.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	18.8	19.7	22.7	25.2	27.2	28.0	28.2	26.8	25.4	22.4	20.6	(27)
(28)			MCD B	19.9	20.8	21.9	24.7	28.8	30.4	31.1	31.6	28.8	27.2	24.0	21.6	(28)
(29)		2%	WB	17.0	17.0	18.5	21.3	24.2	26.5	27.4	27.4	25.9	24.3	20.9	19.0	(29)
(30)			MCD B	18.4	18.3	21.0	23.8	27.0	29.7	30.5	30.4	28.6	26.5	22.3	20.1	(30)
(31)		5%	WB	15.9	15.7	17.6	20.3	23.5	25.8	26.8	26.8	25.4	23.4	19.6	17.7	(31)
(32)			MCD B	17.3	17.2	19.6	22.7	26.2	28.9	30.0	29.9	28.4	25.8	21.3	18.9	(32)
(33)		10%	WB	14.5	14.3	16.6	19.5	22.8	25.3	26.3	26.2	24.7	22.4	18.4	16.5	(33)
(34)			MCD B	15.9	16.0	18.7	21.8	25.5	28.4	29.5	29.3	27.7	24.8	20.3	17.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.0	10.0	9.4	8.3	7.3	7.1	7.0	7.9	9.8	10.9	10.0	(35)
(36)			MCDBR	10.4	10.6	10.4	9.5	8.5	8.0	8.0	7.7	8.3	9.5	10.8	9.9	(36)
(37)			MCWBR	8.2	7.3	6.1	5.5	4.3	4.0	4.0	3.7	3.9	5.2	7.2	7.6	(37)
(38)		5% WB	MCD BR	9.2	9.2	8.9	7.5	6.9	6.9	7.3	6.8	6.7	7.1	8.6	8.9	(38)
(39)			MCWBR	8.1	7.6	6.3	4.8	3.9	3.9	4.2	3.7	3.7	4.5	6.8	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.335	0.367	0.398	0.458	0.484	0.525	0.502	0.429	0.393	0.352	0.331	(40)	
(41)		taud	2.518	2.492	2.427	2.324	2.211	2.185	2.086	2.148	2.344	2.414	2.485	2.537	(41)	
(42)		Ebn at Noon	887	916	909	890	836	811	775	787	834	838	848	860	(42)	
(43)		Edh at Noon	84	95	110	127	144	148	162	150	118	102	86	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.73	3.40	4.55	5.79	6.30	6.50	6.17	5.55	4.72	3.96	3.04	2.43	(44)	
(45)		RadStd	0.17	0.33	0.41	0.38	0.43	0.40	0.38	0.31	0.35	0.42	0.23	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+160
(47) Regional (48 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+163	+118

Nomenclature: See separate page